



2024 Regional Transmission Plan: Appendix_P_2024_RTP_Economic_Analysis

1. 2024 RTP Economic Project Evaluation Results

This section describes the detailed results of the evaluation of the selected transmission project improvement based on Production Cost Savings (PCS) Test, Generator Revenue Reduction (GRR) Test, and Total Consumer Energy Cost Reduction (TCECR) Test.

Richland – Wortham Junction 69-kV Line Upgrade (Project 1)

The Richland to Wortham Junction 69-kV line was congested under the contingency loss of the 138-kV line from Groesbeck to Fewell. Table 1 includes details of the congestion on this line.

Table 1: Richland to Wortham Junction 69-kV Congestion Details

Congested Element	Richland to Wortham Junction 69-kV line			
Most Severe Contingency	Loss of Groesbeck to Fewell 138-kV line			
	Capacity (MVA)	Study Year	% of Time Congested	Congestion Rent (\$M)
	41	2026	2.84	27.2
		2029	3.54	34.2

Upgrading the Richland to Wortham Junction line alleviated the congestion in the area without creating any additional congestion.

The project was evaluated in both the 2026 and 2029 economic cases, and the economic analysis results are included in Table 2. The project demonstrated an annual decrease of \$1.08 million in production cost and the break-even capital cost is \$8.4 million. Additionally, the project resulted in an annual consumer energy cost savings of \$25.0 million with a break-even cost of \$198.5 million. On the other hand, the project caused an increase of \$0.3 million in the generator annual revenue.

Table 2: Economic Analysis Results for Richland to Wortham Junction 69-kV Line Upgrade

	Adj. Savings in 2026 (M\$)	Adj. Savings in 2029 (M\$)	Levelized Savings in 2026 Dollars (\$M)	Break-even Capital Cost in 2026 Dollars (\$M)
PCS Test	1.3	1.0	1.08	8.4
GRR Test	3.4	-4.3	-0.3	N/A
TCECR Test	25.9	27.7	25.0	198.5

Farmland Area 345-kV Upgrade (Project 2)

The 345-kV line from Farmland to Long Draw was congested under the contingency loss of the 345-kV double-circuit line from White River to Cottonwood. Table 3 provides details of the congestion on this line.

Table 3: Farmland to Long Draw 345-kV Line Congestion Details

Congested Element	Farmland to Long Draw 345-kV line			
Most Severe Contingency	Loss of White River to Cottonwood 345-kV double circuit line			
	Capacity (MVA)	Study Year	% of Time Congested	Congestion Rent (\$M)
	1,223	2026	8.95	19.0
		2029	16.54	63.6

The project encompasses the upgrade of the 345-kV lines from Fiddlewood to Farmland to Long Draw. The upgrade of the lines was evaluated in the 2029 economic case, and the economic analysis results are included in Table 4. The project resulted in an annual production cost savings of \$5.54 million with a break-even cost of \$42.9 million. Additionally, the project resulted in an annual consumer energy cost savings of \$27.2 million with a break-even cost of \$216 million. However, the project increased the generator revenue by \$5.1 million.

Table 4: Economic Analysis Results for the Farmland 345-kV Area Upgrade

	Adj. Savings in 2029 (M\$)	Break-even Capital Cost in 2029 Dollars (\$M)
PCS Test	5.5	42.9
GRR Test	-5.1	N/A
TCECR Test	27.2	216

Lubbock Area Improvement (Project 3)

This project aimed at addressing the congestion on the MacKenzie Substation to Northeast Substation 115-kV line. Details of this congestion can be found in Table 5.

Table 5: MacKenzie Substation to Northeast Substation 115-kV Line Congestion Details

Congested Element	MacKenzie Substation to Northeast Substation 115-kV line			
Most Severe Contingency	Loss of Blackwater Draw Switch to Double Mountain Switch 345-kV line			
	Capacity (MVA)	Study Year	% of Time Congested	Congestion Rent (\$M)
	218	2026	2.33	14.8
		2029	22.59	181

Upgrading the 115-kV lines connecting the McDonald to Northwest to Yellow House Canyon to MacKenzie to Northeast to Dunbar to Holly substations, with a total of approximately 22.6 miles, was evaluated, in the 2029 economic case, to alleviate the congestions observed in the Lubbock area. The economic analysis results are included in Table 6. The project reduced the production cost by \$0.5 million with a break-even capital cost of \$3.9 million while also decreasing the generator revenue by

\$3.57 million with a break-even capital cost of \$28.4 million. Additionally, the project resulted in an annual consumer energy cost savings of \$35.9 million with a break-even cost of \$285 million.

Table 6: Economic Analysis Details for Lubbock Area Upgrade

	Adj. Savings in 2029 (M\$)	Break-even Capital Cost in 2029 Dollars (\$M)
PCS Test	0.5	3.9
GRR Test	3.6	28.4
TCECR Test	35.9	285

Coast Weather Zone Improvement Option 1 (Project 4)

Heavy congestions have been observed on three 345-kV lines in the Coast Weather Zone, i.e., PH Robinson to Meadow, Refuge to Jones Creek, and Zenith to TH Wharton. Details of these congestions can be found in Table 7.

Table 7: 345-kV Lines Congestions in Coast Weather Zone

Congested Element	PH Robinson to Meadow 345-kV line			
Most Severe Contingency	N/A ¹			
	Capacity (MVA)	Study Year	% of Time Congested	Congestion Rent (\$M)
	1,195	2026	20.34	53.6
		2029	15.67	41.8

Congested Element	Refuge to Jones Creek 345-kV line			
Most Severe Contingency	Loss of Bailey to Blue and Wolf to Hillje 345-kV Lines			
	Capacity (MVA)	Study Year	% of Time Congested	Congestion Rent (\$M)
	1,450	2026	6.84	21.9
		2029	6.34	23.4

Congested Element	Zenith to TH Wharton (Circuit 1) 345-kV line			
Most Severe Contingency	Loss of Zenith to TH Wharton (Circuit 2) 345-kV Line			
	Capacity (MVA)	Study Year	% of Time Congested	Congestion Rent (\$M)
	1,314	2026	2.36	9.7
		2029	1.92	9.8

¹ Base case (N-0) violation

New 345-kV lines from South Texas Project (STP) to Bailey to PH Robinson were evaluated, to relieve the congestions observed. Economic analysis results for both the 2026 and 2029 study years are included in Table 8. The evaluation results showed that the annual production cost was reduced by \$6.59 million, leading to a break-even capital cost of \$51.1 million. However, the project resulted in an increase in generator revenue and consumer energy cost of \$132.6 million and \$107.1 million, respectively.

Table 8: Economic Analysis Results for Coast Weather Zone Improvement Option 1

	Adj. Savings in 2026 (M\$)	Adj. Savings in 2029 (M\$)	Levelized Savings in 2026 Dollars (\$M)	Break-even Capital Cost in 2026 Dollars (\$M)
PCS Test	8.2	5.9	6.6	51.1
GRR Test	-124.3	-160.9	-132.6	N/A
TCECR Test	-96.6	134	-107.1	N/A

Coast Weather Zone Improvement Option 2 (Project 5)

To relieve these congestions observed in Coast Weather Zone, three new 345-kV lines were evaluated, i.e., South Texas Project to Bailey, Bailey to Ph Robinson, and Bailey to Zenith as well as upgrading the Zenith to TH Wharton 345-kV double-circuit line and the PH Robinson to Meadow 345-kV double-circuit line. The project was evaluated in both the 2026 and 2029 economic cases, and the economic analysis results are included in Table 9. The project reduced the production cost by \$7.85 million with a break-even capital cost of \$60.9 million. However, the project resulted in an increase in generator revenue and consumer energy cost of \$144.7 million and \$113 million, respectively.

Table 9: Economic Analysis Results for Coast Weather Zone Improvement Option 2

	Adj. Savings in 2026 (M\$)	Adj. Savings in 2029 (M\$)	Levelized Savings in 2026 Dollars (\$M)	Break-even Capital Cost in 2026 Dollars (\$M)
PCS Test	11.6	5.1	7.9	60.9
GRR Test	-138.0	-173.2	-144.7	N/A
CCS Test	-101	-143	-113	N/A

Murray – Paint Creek 138-kV Line Upgrade (Project 6)

The 138-kV line from Murray to Paint Creek was congested under the contingency loss of Tonkawa Switch to Graham SES 345-kV line, Graham SES to Long Creek 345-kV line, Graham SES to Cook Field Road Switch 345-kV line, China Grove to Dunn Pod 138-kV line, Radium to Paint Creek 138-kV line, and Radium to Dunn Pod 138-kV line. Table 10 shows the details of the observed congestion.

Table 10: Murray to Paint Creek 138-kV Line Congestion Detail

Congested Element	Murray to Paint Creek 138-kV line
Most Severe Contingency	Loss of Tonkawa Switch to Graham SES 345-kV Line, Graham SES to Long Creek 345-kV Line, Graham SES to Cook Field Road Switch 345-kV Line,

	China Grove to Dunn Pod 138-kV Line, Radium to Paint Creek 138-kV Line, and Radium to Dunn Pod 138-kV Line			
	Capacity (MVA)	Study Year	% of Time Congested	Congestion Rent (\$M)
	143	2026	0.13	0.1
		2029	2.51	18.3

The proposed project was then evaluated in the 2029 economic case, and the economic analysis results are included in Table 11. The project demonstrated an annual decrease of \$2.45 million in production cost and the break-even capital cost is \$19.01 million. The project also caused a reduction of \$3.9 million in the generator annual revenue with a break-even capital cost of \$31 million. Additionally, the project resulted in an annual consumer energy cost savings of \$23.6 million with a break-even cost of \$187.6 million.

Table 11: Economic Analysis Results for Murray to Paint Creek 138-kV Line Upgrade

	Adj. Savings in 2029 (M\$)	Break-even Capital Cost in 2029 Dollars (\$M)
PCS Test	2.5	19.0
GRR Test	3.9	31.0
TCECR Test	23.6	187.6

Lon Hill – Whitepoint 345-kV Line Upgrade (Project 7)

The 345-kV line from Lon Hill to Whitepoint was congested under the contingency loss of the 345-kV line from Angstrom to Grissom. Table 12 shows the details of the observed congestion.

Table 12: Lon Hill to Whitepoint 345-kV Line Congestion Detail

Congested Element	Lon Hill to Whitepoint 345-kV line			
Most Severe Contingency	Loss of Angstrom to Grissom 345-kV line			
	Capacity (MVA)	Study Year	% of Time Congested	Congestion Rent (\$M)
	1,057	2026	13.6	30.9
		2029	8.2	17.4

To relieve the congestion, the 345-kV line from Lon Hill to Whitepoint was upgraded, totaling approximately 20.5 miles. The project was evaluated in both the 2026 and 2029 economic cases, and the economic analysis results are included in Table 13. The project reduced the production cost by \$0.62 million with a break-even capital cost of \$4.8 million while also reducing the generator revenue by \$6.9 million with a break-even capital cost of \$55 million. Additionally, the project resulted in an annual consumer energy cost savings of \$17 million with a break-even cost of \$134.6 million.

Table 13: Economic Analysis Results for Lon Hill to Whitepoint 345-kV Line Upgrade

	Adj. Savings in 2026 (M\$)	Adj. Savings in 2029 (M\$)	Levelized Savings in 2026 Dollars (\$M)	Break-even Capital Cost in 2026 Dollars (\$M)
PCS Test	2.1	-0.8	0.6	4.8
GRR Test	18.0	-3.8	6.9	55.0
TCECR Test	29.9	5.8	17	134.9

Bell County East Switch – Scooter 345-kV Upgrade (Project 8)

The 345-kV line from Bell County East Switch to Scooter was congested under the contingency loss of the parallel 345-kV line from Bell County East Switch to Scooter. Table 14 shows the details of the observed congestion.

Table 14: Bell County East Switch to Scooter 345-kV Line Congestion Detail

Congested Element	Bell County East Switch to Scooter 345-kV line			
Most Severe Contingency	Loss of parallel Bell County East Switch to Scooter 345-kV line			
	Capacity (MVA)	Study Year	% of Time Congested	Congestion Rent (\$M)
	1,072	2026	20.61	38.5
		2029	16.39	33.1

To relieve the congestion, the 345-kV double-circuit line from Bell County East Switch to Scooter was upgraded, totaling approximately 59.4 miles. The project was evaluated in the 2029 economic case, and the economic analysis results are included in Table 15. The project reduced the production cost by \$2.4 million with a break-even capital cost of \$18.3 million while increasing the generator revenue by \$21.6 million. Additionally, the project resulted in an annual consumer energy cost increase of \$13.7 million.

Table 15: Economic Analysis Results for Bell County East Switch to Scooter 345-kV Line Upgrade

	Adj. Savings in 2029 (M\$)	Break-even Capital Cost in 2029 Dollars (\$M)
PCS Test	2.4	18.3
GRR Test	-21.6	N/A
TCECR Test	-13.7	N/A

2. Economic Analysis Result Summary

The economic benefits for 2024 RTP economic projects were calculated for the PCS Test, GRR Test, and TCECR Test, as shown in Table 16. The break-even capital cost for 2024 RTP economic projects, if applicable, is provided in Table 17. All cost savings or the break-even capital cost is in 2026 dollars except designated as (*) is in 2029 dollars.

Table 16: Economic Analysis Results for 2024 RTP Economic Project Evaluation

Index	Description	Production Cost Savings (\$M)	Generator Revenue Reduction (\$M)	Total Consumer Energy Cost Reduction (\$M)
Project 1	Richland to Wortham 69-kV Upgrade	\$1.08	-	\$25.00
Project 2 (*)	Farmland Area Improvement (Upgrade Farmland – Long Draw and Farmland – Fiddlewood Switch 345-kV lines)	\$5.54	-	\$27.23
Project 3 (*)	Lubbock Area Improvement (Upgrade the following 115-kV transmission lines: Yellow House – Northwest – McDonald – Mackenzie – Northeast – Dunbar – Holly)	\$0.50	\$3.57	\$35.91
Project 4	Coast Weather Zone Improvement Option 1 (New 345-kV line STP – Bailey New 345-kV line Bailey – PH Robinson)	\$6.59	-	-
Project 5	Coast Weather Zone Improvement Option 2 (Project 4 Plus New 345-kV line Bailey – Zenith Upgrade Zenith – TH Wharton DCKT 345-kV Upgrade PH Robinson – Meadow 345-kV line)	\$7.85	-	-
Project 6 (*)	Murray – Paint Creek 138-kV Upgrade	\$2.45	\$3.90	\$23.63
Project 7	Lon Hill – White Point 345-kV Upgrade	\$0.62	\$6.93	\$17.00
Project 8 (*)	Bell County East Switch – Scooter 345-kV Upgrade	\$2.36	-	-

Table 17: Break-even Capital Cost for 2024 RTP Economic Project Evaluation

Index	Description	PCS Test Break-Even Capital Cost (\$M)	GRR Test Break-Even Capital Cost (\$M)	TCECR Test Break-Even Capital Cost (\$M)
Project 1	Richland to Wortham 69-kV Upgrade	\$ 8.40	-	\$ 198.45
Project 2 (*)	Farmland Area Improvement (Upgrade Farmland – Long Draw and Farmland – Fiddlewood Switch 345-kV lines)	\$ 42.93	-	\$ 216.09
Project 3 (*)	Lubbock Area Improvement (Upgrade the following 115-kV transmission lines: Yellow House – Northwest – McDonald – Mackenzie – Northeast – Dunbar – Holly)	\$ 3.91	\$ 28.37	\$ 285.01
Project 4	Coast Weather Zone Improvement Option 1 (New 345- kV line STP – Bailey New 345-kV line Bailey – PH Robinson)	\$ 51.06	-	-
Project 5	Coast Weather Zone Improvement Option 2 (Project 4 Plus New 345-kV line Bailey – Zenith Upgrade Zenith – TH Wharton DCKT 345-kV Upgrade PH Robinson – Meadow 345-kV line)	\$ 60.87	-	-
Project 6 (*)	Murray – Paint Creek 138-kV Upgrade	\$ 19.01	\$ 30.97	\$ 87.55
Project 7	Lon Hill – White Point 345-kV Upgrade	\$ 4.80	\$ 55.02	\$ 134.91
Project 8 (*)	Bell County East Switch – Scooter 345-kV Upgrade	\$ 18.33	-	-